

Comparsion_Graph

March 6, 2026

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[1]: import pandas as pd
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[2]: import matplotlib.pyplot as plt
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[3]: import numpy as np
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[4]: orig = pd.read_csv("original_hula_results.csv")
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[5]: mod = pd.read_csv("modified_hula_results.csv")
```

```
[6]: print("Original shape:", orig.shape)
```

Original shape: (1000, 5)

```
[7]: print("Modified shape:", mod.shape)
```

Modified shape: (1000, 5)

```
[8]: orig.head()
```

```
[8]:
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	load_percent	time_sec	probes_per_sec	total_probes	overhead
0	10	0	5.254	5.25	0.000027
1	10	1	5.005	10.26	0.000026
2	10	2	4.882	15.14	0.000025
3	10	3	5.000	20.14	0.000026
4	10	4	5.153	25.29	0.000026

```
[9]: orig_10 = orig[orig["load_percent"] == 10]  
mod_10 = mod[mod["load_percent"] == 10]
```

```
[10]: plt.figure()  
plt.plot(orig_10["time_sec"], orig_10["probes_per_sec"], label="Original HULA")  
plt.plot(mod_10["time_sec"], mod_10["probes_per_sec"], label="Modified HULA")  
  
plt.xlabel("Time (seconds)")  
plt.ylabel("Probes per Second")  
plt.title("Time vs Probe Rate (10% Load)")  
plt.legend()
```

```
plt.grid(True)

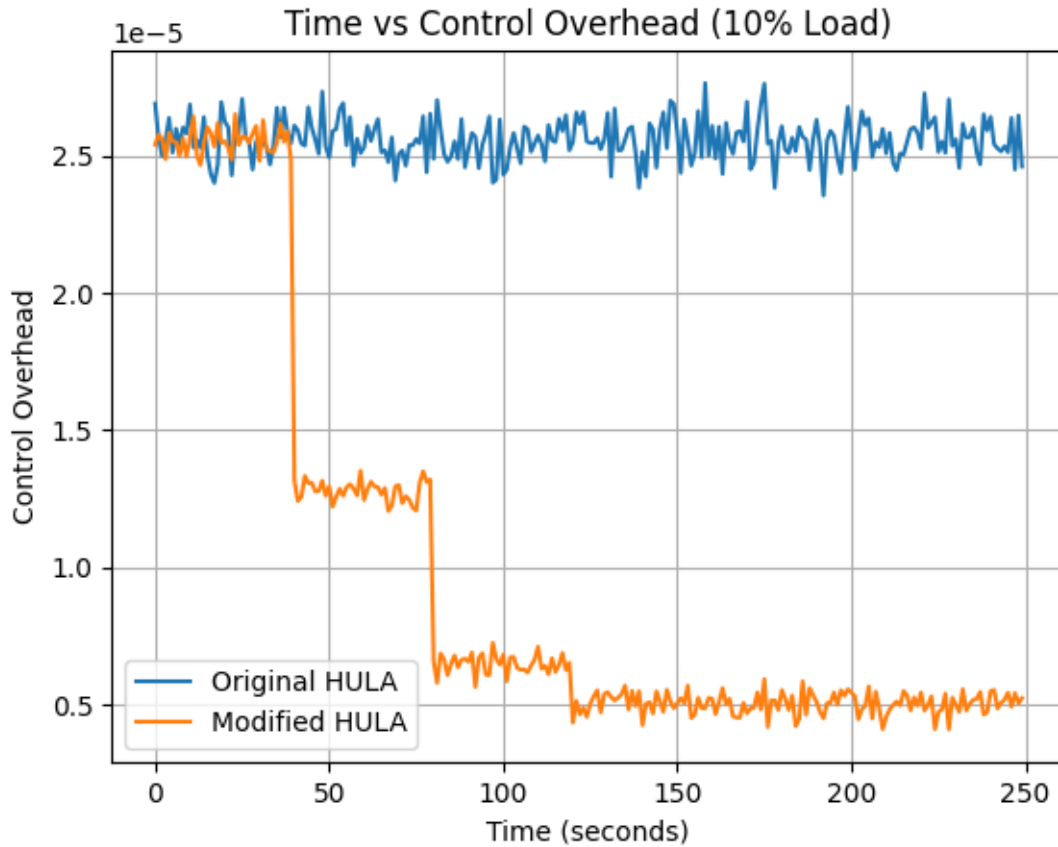
plt.show()
```



```
[11]: plt.figure()
plt.plot(orig_10["time_sec"], orig_10["overhead"], label="Original HULA")
plt.plot(mod_10["time_sec"], mod_10["overhead"], label="Modified HULA")

plt.xlabel("Time (seconds)")
plt.ylabel("Control Overhead")
plt.title("Time vs Control Overhead (10% Load)")
plt.legend()
plt.grid(True)

plt.show()
```

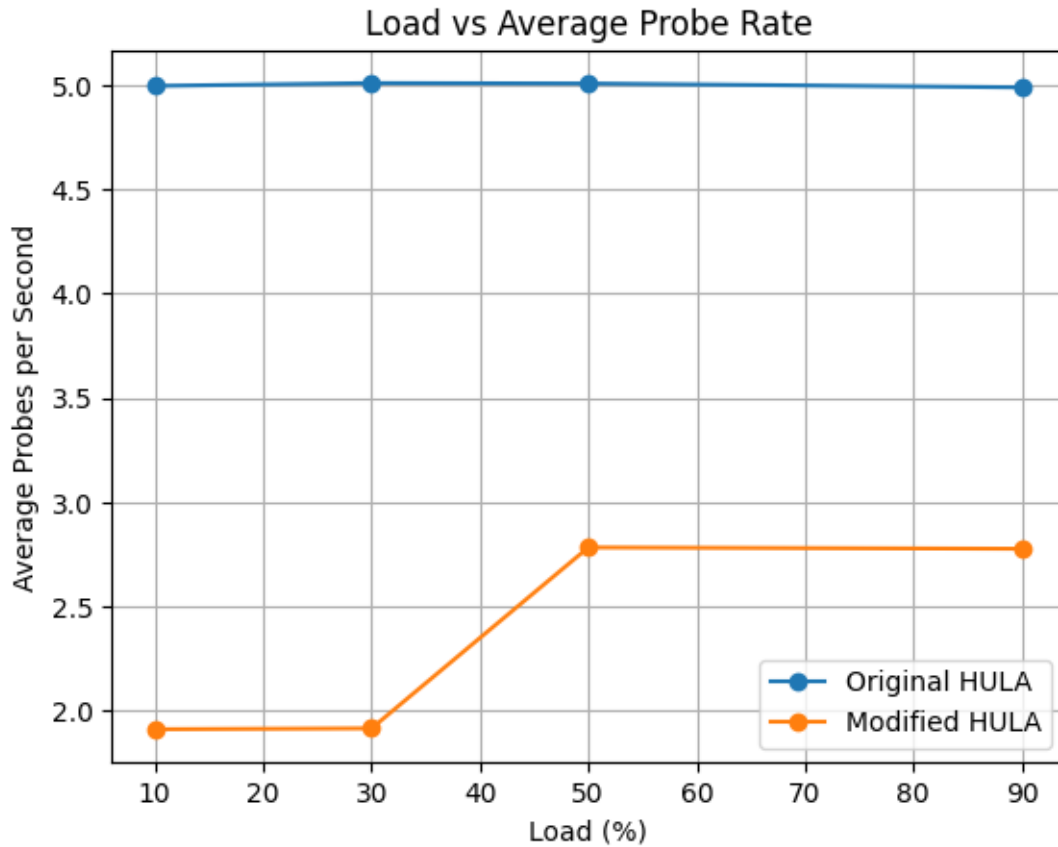


```
[12]: orig_avg = orig.groupby("load_percent")["probes_per_sec"].mean()
mod_avg = mod.groupby("load_percent")["probes_per_sec"].mean()

plt.figure()
plt.plot(orig_avg.index, orig_avg.values, marker='o', label="Original HULA")
plt.plot(mod_avg.index, mod_avg.values, marker='o', label="Modified HULA")

plt.xlabel("Load (%)")
plt.ylabel("Average Probes per Second")
plt.title("Load vs Average Probe Rate")
plt.legend()
plt.grid(True)

plt.show()
```

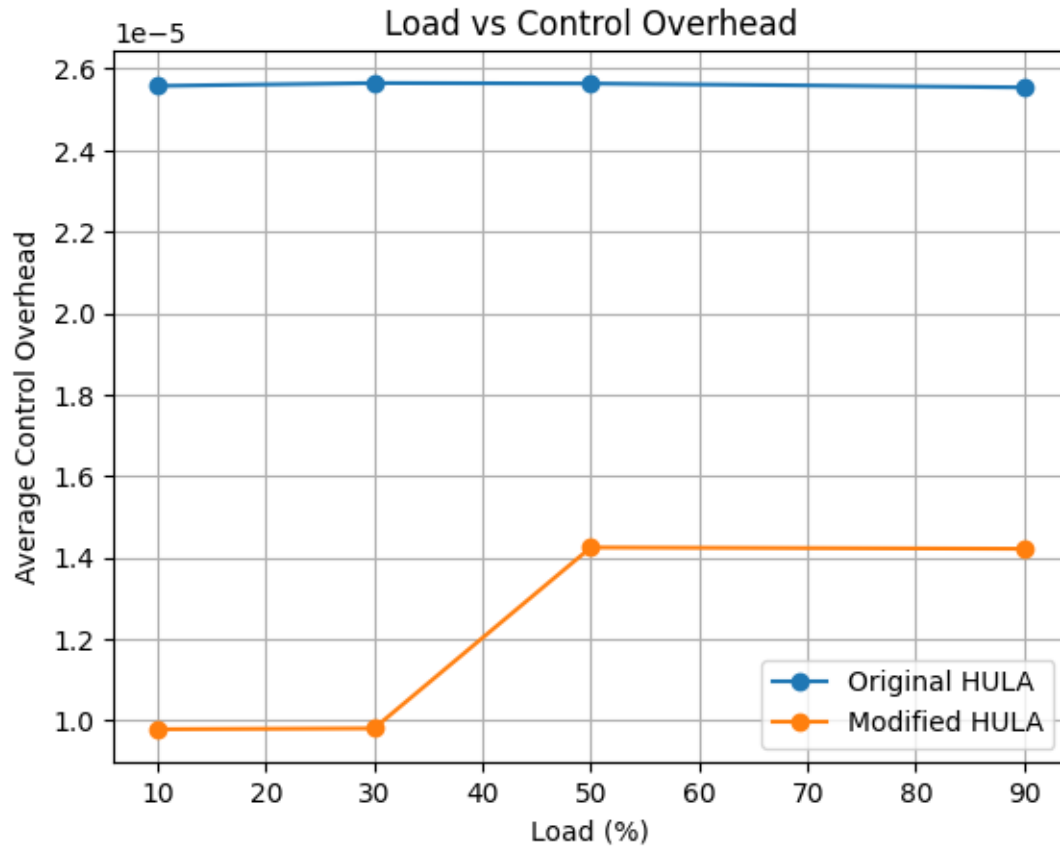


```
[13]: orig_over = orig.groupby("load_percent")["overhead"].mean()
mod_over = mod.groupby("load_percent")["overhead"].mean()

plt.figure()
plt.plot(orig_over.index, orig_over.values, marker='o', label="Original HULA")
plt.plot(mod_over.index, mod_over.values, marker='o', label="Modified HULA")

plt.xlabel("Load (%)")
plt.ylabel("Average Control Overhead")
plt.title("Load vs Control Overhead")
plt.legend()
plt.grid(True)

plt.show()
```



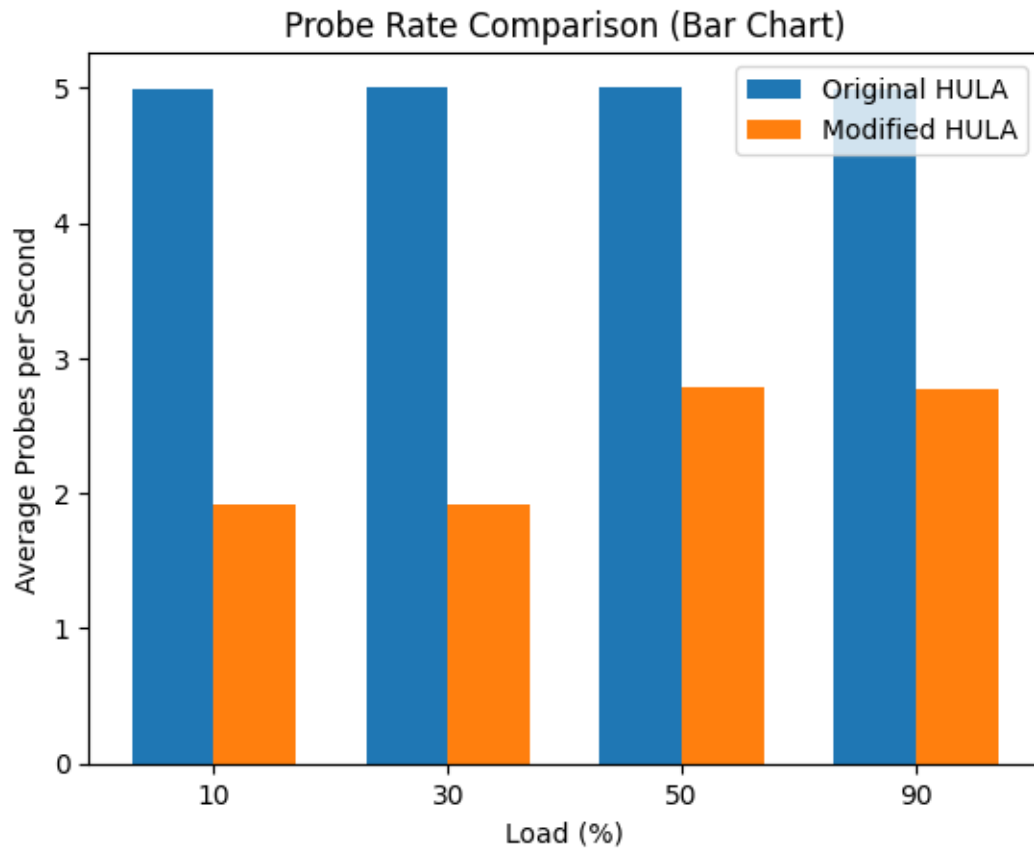
```
[14]: loads = orig_avg.index

x = np.arange(len.loads))
width = 0.35

plt.figure()
plt.bar(x - width/2, orig_avg.values, width, label="Original HULA")
plt.bar(x + width/2, mod_avg.values, width, label="Modified HULA")

plt.xlabel("Load (%)")
plt.ylabel("Average Probes per Second")
plt.title("Probe Rate Comparison (Bar Chart)")
plt.xticks(x, loads)
plt.legend()

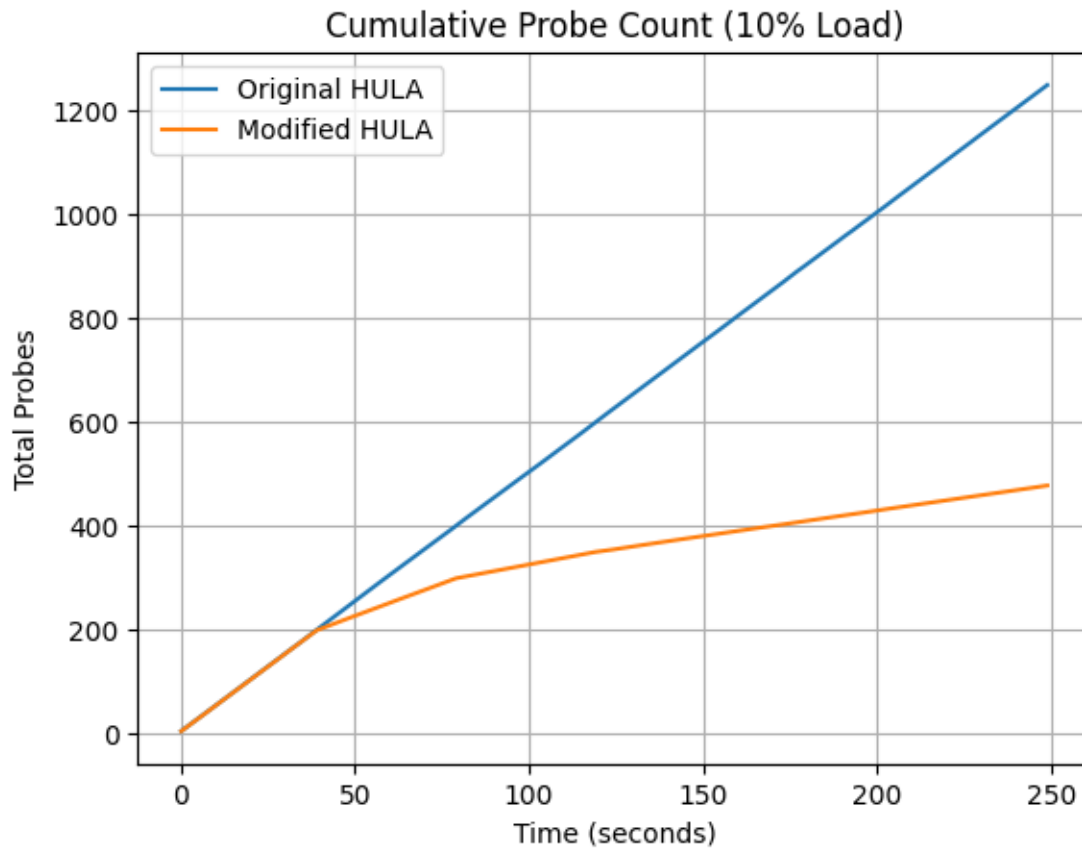
plt.show()
```



```
[15]: plt.figure()
plt.plot(orig_10["time_sec"], orig_10["total_probes"], label="Original HULA")
plt.plot(mod_10["time_sec"], mod_10["total_probes"], label="Modified HULA")

plt.xlabel("Time (seconds)")
plt.ylabel("Total Probes")
plt.title("Cumulative Probe Count (10% Load)")
plt.legend()
plt.grid(True)

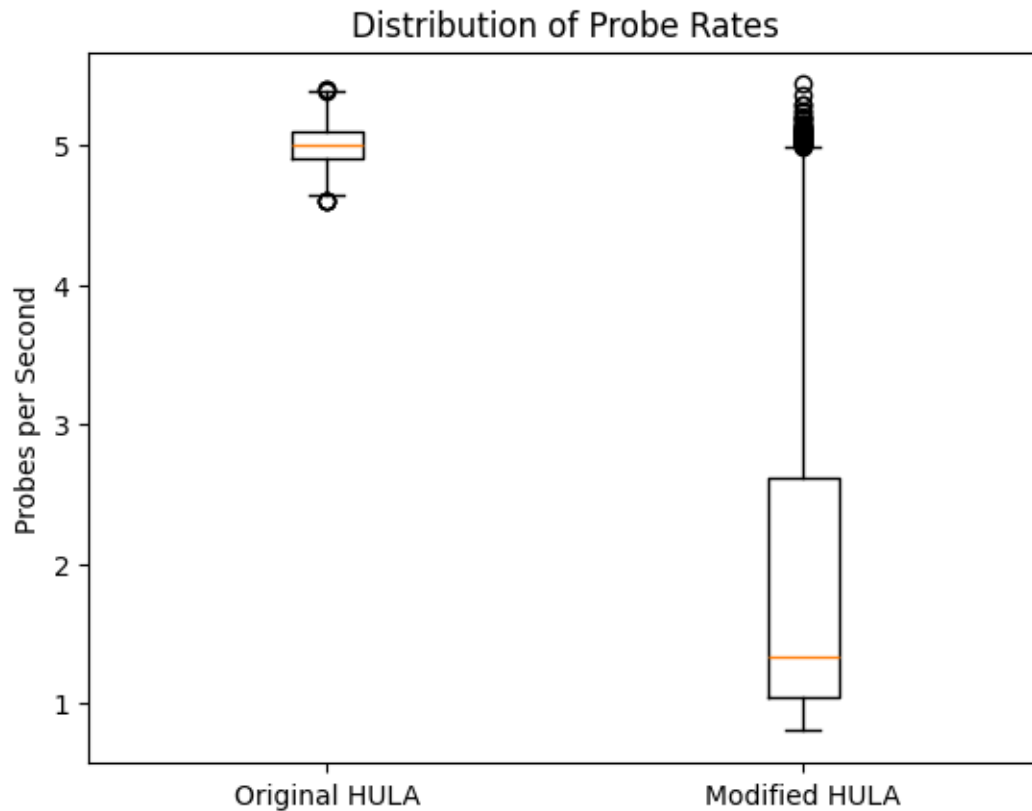
plt.show()
```



```
[16]: plt.figure()
plt.boxplot([
    orig["probes_per_sec"],
    mod["probes_per_sec"]
])

plt.xticks([1, 2], ["Original HULA", "Modified HULA"])
plt.ylabel("Probes per Second")
plt.title("Distribution of Probe Rates")

plt.show()
```



```
[17]: time = np.arange(0, 120)
```

```
[18]: mod_probe = []
      current_rate = 5

      for t in time:

          # Congestion spike between 40s and 70s
          if 40 <= t <= 70:
              current_rate = 5    # reset to high rate during congestion

          else:
              # Exponential backoff during stability
              if t % 10 == 0 and current_rate > 1:
                  current_rate = current_rate / 2

              # Minimum probe rate = 1
              if current_rate < 1:
                  current_rate = 1

      mod_probe.append(current_rate)
```

```
[19]: orig_probe = np.ones_like(time) * 5
```

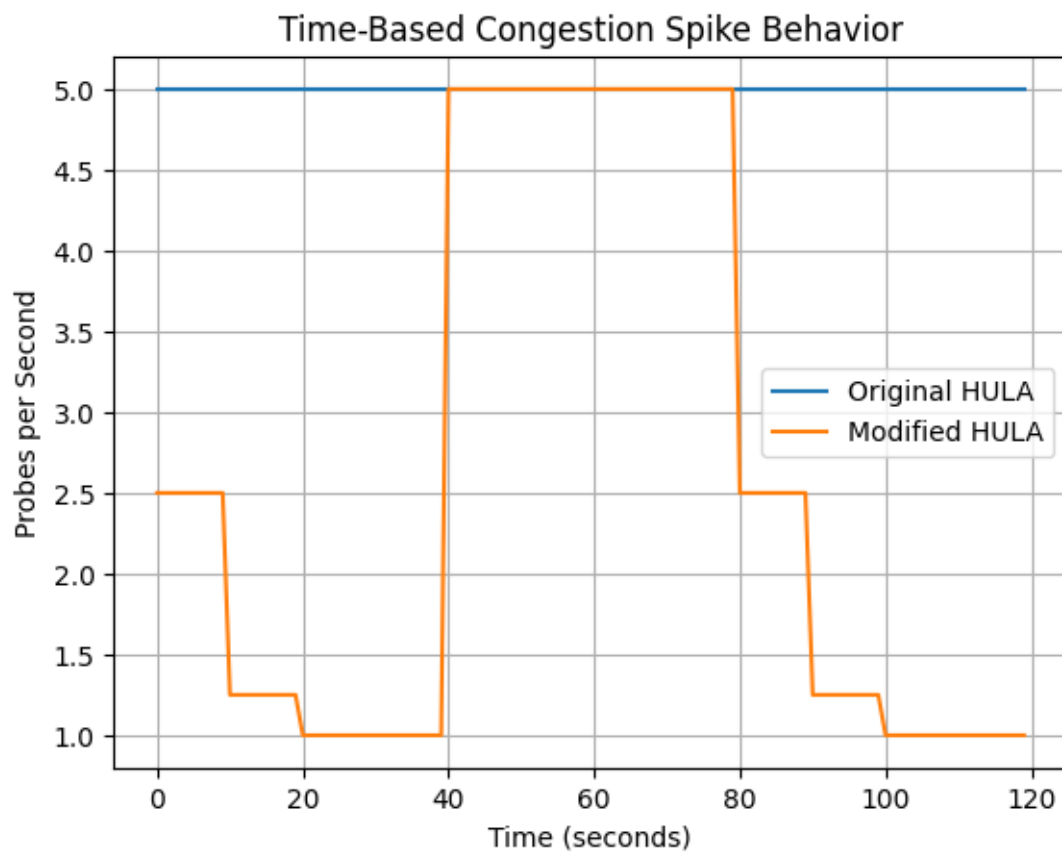
```
[20]: plt.figure()

plt.plot(time, orig_probe, label="Original HULA")
plt.plot(time, mod_probe, label="Modified HULA")

plt.xlabel("Time (seconds)")
plt.ylabel("Probes per Second")
plt.title("Time-Based Congestion Spike Behavior")

plt.legend()
plt.grid(True)

plt.show()
```



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[ ]:
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